Master’s Degree in Data Science
Academic Year: 2025-2026

HOMEWORK 2

CUSTOMER LIFETIME VALUE (CLV) ANALYSIS

MARKETING ANALYTICS

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EXECUTIVE SUMMARY

Objective

The goal of this project is to transform raw customer data into actionable managerial insights in order to optimize marketing investments and customer profitability for an Indian Fast-Moving Consumer Goods (FMCG) retailer.

Business Scenario

Acting as the marketing analytics team of an Indian FMCG retailer, this analysis addresses specific challenges expressed by senior management, such as:

- Identifying high-value segments to maximize overall revenue and strengthen long-term relationships.
- Recognizing at-risk customers to develop win-back initiatives.
- Developing tiered loyalty programs through exclusive benefits and personalized offers.
- Defining optimal resource allocation strategies to balance acquisition and retention strategies.

Methodology

To address management's strategic questions, the developed framework consists of multiple phases:

- **Data preparation and exploratory data analysis**: implementation of preprocessing steps and discovery of behavioural patterns and purchasing trends.
- **RFM analysis**: categorization of the customer base based on recency, frequency, and monetary value of historical transactions into 5 levels per each metric.
- **Predictive CLV modelling and segmentation**: estimation of the predicted value of customers over the next 6 months to categorize them into 4 layers.
- **Managerial recommendations**: translation of analytical results into actionable managerial insights.

Key Insights

The top 20% of customers generate approximately 50% of total revenue. Beverages is the primary category for recurring purchases, maintaining customer engagement. Resource allocation should prioritize high-value customer retention and re-engagement for low-value segments.

DATA OVERVIEW

(SEE APPENDIX A FOR DATA DICTIONARY AND DISCUSSION)

Dataset

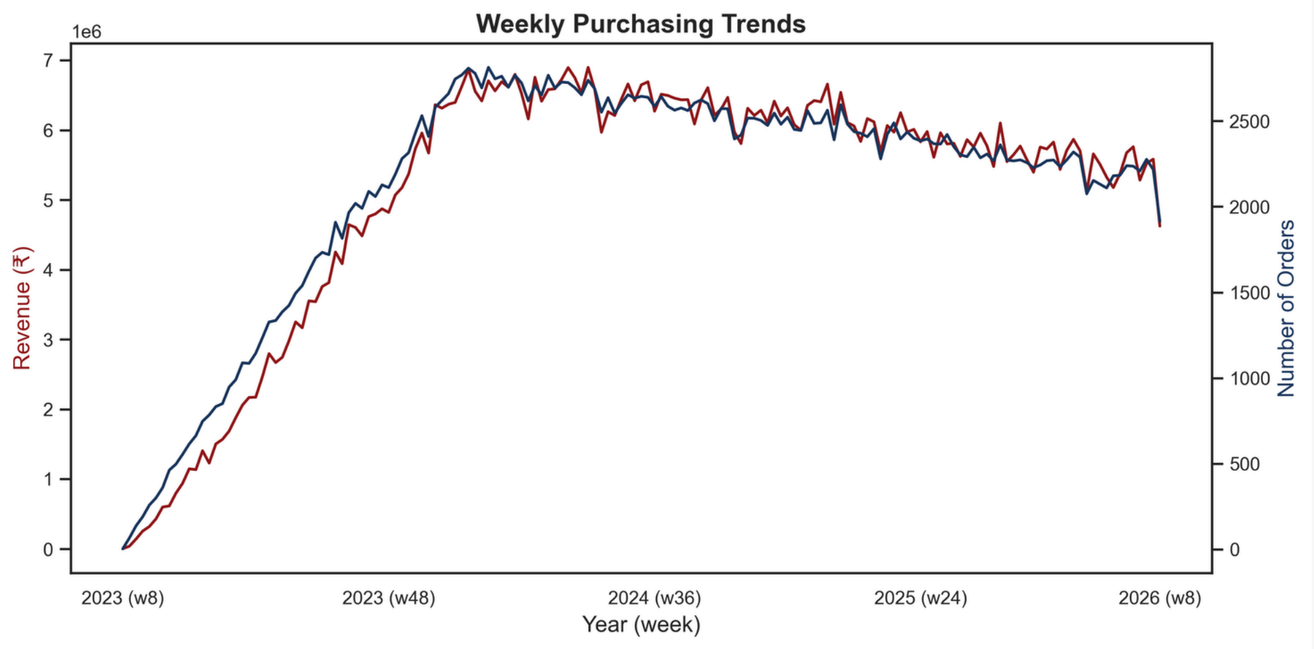
- Structure: 14 relational tables (Indian FMCG retailer).
- Volume: 356,242 customer orders across 10 product categories.
- Time horizon: February 2023 - February 2026.

Data Quality and Preprocessing

- Missing values: 70% missing in the variable promo_id of the table order_item, interpreted as a valid condition and stored as NULL.
- Temporal columns: converted date and time variables to proper data types.
- Discount inconsistency: capped item_discount at maximum line revenue (unit_price x quantity) to prevent negative transaction values.
- Product pricing: mapped 2.57% of unpriced order items through a forward approach.
- Referential integrity: aligned campaign_touch and marketing_channel tables.

EXPLORATORY ANALYSIS

(SEE APPENDIX B FOR FURTHER VISUALIZATIONS)



From February 2023, the company experienced a strong growth phase, with both revenue and order volume increasing steadily throughout the year. In particular, weekly revenue reached approximately ₹ 7,000,000 by the end of 2023, while weekly orders exceeded 2,500. Although short-term fluctuations can be observed, probably due to promotional campaigns or seasonal effects, the overall trend remained positive. Starting in 2024, both metrics began to decrease gradually before stabilising at around ₹ 5,000,000 in weekly revenue and approximately 2,000 weekly orders during the most recent period.

RFM ANALYSIS

(SEE APPENDIX C FOR CUSTOMER TABLE AND ADDITIONAL VISUALIZATIONS)

RFM Metrics

Recency, frequency, and monetary are three widely used customer attributes for behavioural analysis and segmentation. In particular:

- **Recency**: number of days since the customer's most recent purchase.
- **Frequency**: total number of purchases made by the customer in the analysed period.
- **Monetary value**: total amount of money spent by the customer over the considered period.

RFM Scoring

- Assign ordinal scores from 1 to 5, where higher values indicate more valuable customers, to each continuous metric by using quantiles, so that the number of customers in each group is the same.
- Compute the final score as:

R x 100 + F x 50 + M.

Scoring Analysis

The summary statistics highlight clear differences across the three RFM dimensions. For recency and frequency, customers with the highest score are recent buyers and typically make a large number of purchases, while those with the lowest score have not purchased for a long time and show only a few purchases. In contrast, monetary value presents more significant differences, indicating the presence of a small group of high-value customers.

Recency	Min	Mean	Max
1	261	577.41	1,068
2	56	124.83	261
3	21	34.98	56
4	7	13.5	21
5	0	3.07	7

Frequency	Min	Mean	Max
1	1	5.97	10
2	10	14.62	19
3	19	24.76	31
4	31	40.01	52
5	52	92.47	846

Monetary	Min	Mean	Max
1	34.21	13,516.80	25,059.08
2	25,064.45	36,267.17	47,760.45
3	47,789.71	61,301.68	77,432.03
4	77,446.86	100,295.79	129,520.81
5	129,530.06	230,583.77	2,201,660.47

CLV ANALYSIS

(SEE APPENDIX D FOR CUSTOMER TABLE)

Customer Lifetime Value (CLV) plays a crucial role in the estimation of the profit that a business can expect to generate from its relationship with a certain customer over time.

Customers	Revenue	Transactions
9,378	₹ 828,990,161.5	333,571

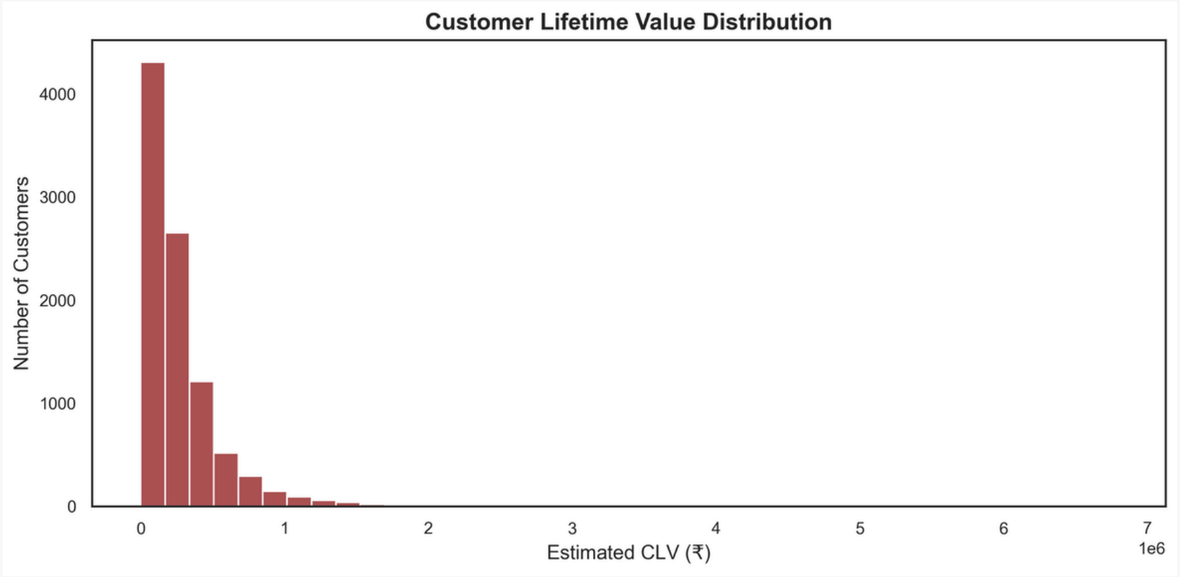
CLV Metrics

- **Average Purchase Value**: APV = Revenue / Transactions
- **Purchase Frequency**: PF = Transactions / Customers
- **Churn Rate**: CR = Churned Customers / Total Customers

CLV Scoring

Final score: **CLV = (APV x PF) / CR**.

APV	PF	CR	CLV
₹ 2,485.2	35.57	32.45%	₹ 272,425.29



CLV varies considerably across the customer base, ranging from just over ₹ 100 to more than ₹ 6.7 million. The average CLV is approximately ₹ 272,425, while 75% of customers remain below ₹ 346,688. Therefore, a limited number of customers contribute significantly to the overall revenue of the business.

CLV PREDICTION

(SEE APPENDIX E FOR CUSTOMER TABLE AND ADDITIONAL VISUALIZATIONS)

Methodology and Justification

A probabilistic framework combining the Beta Geometric Negative Binomial Distribution (BG/NBD) model and the Gamma-Gamma distribution model is used to predict the Customer Lifetime Value. Unlike classical approaches that only rely on past behaviours, these methods account for the heterogeneity in customer purchasing patterns.

BG/NBD model predicts the expected number of future transactions, considering customer recency and frequency.

r	α	a	b
1.47	34.57	0.04	1.12

Gamma-Gamma model predicts the expected monetary value of future transactions, assuming the independence between the average transaction value and transaction frequency.

corr(monetary, frequency)	0.05

p	q	v
3.07	0.19	3.04

Performance Evaluation

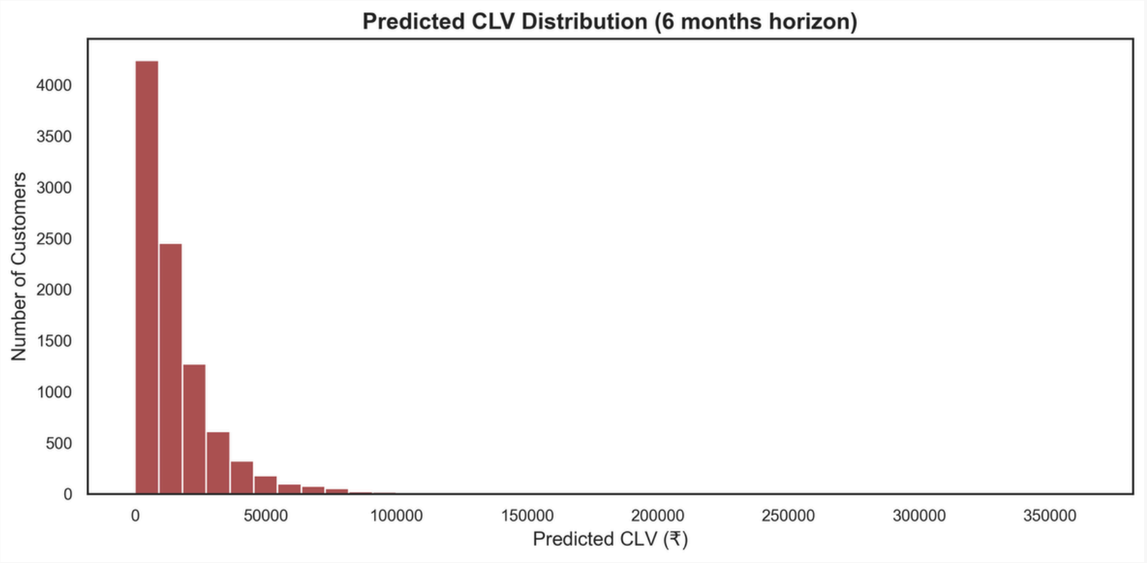
The model's performance is evaluated using a calibration-holdout split and considering some metrics (RMSE, MAE, R^2).

Calibration (75%): from 2023-02-26 to 2025-05-24
Holdout (25%): from 2025-05-25 to 2026-02-21

RMSE	MAE	R^2
15,487.67	10,399.59	0.69

Prediction over a 6-Month Horizon

The distribution of CLV within the six-month prediction horizon reveals significant heterogeneity among customers, with a mean of approximately ₹ 14,965 and a high maximum value, as a limited number of buyers strongly contribute to the expected revenue.

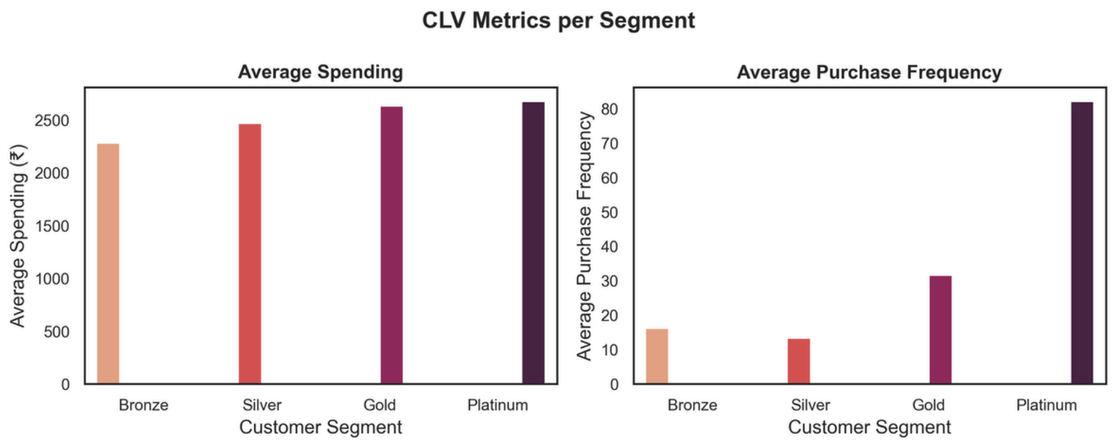


CUSTOMER SEGMENTATION

(SEE APPENDIX F FOR SEGMENT SUMMARY TABLE AND FURTHER VISUALIZATION)

Customers are divided into four groups based on their estimated Customer Lifetime Value:

- **Bronze**: bottom 20% characterized by a few purchases and a very low probability of remaining active.
- **Silver**: next 30% of the customer base with a moderate contribution, while having a high probability of remaining active.
- **Gold**: following 30% of customers who remain engaged and contribute significantly to the overall business revenue.
- **Platinum**: top 20% composed of the most valuable customers, showing the greatest revenue potential.



RECOMMENDATIONS

(SEE APPENDIX G FOR COMPLETE DISCUSSION)

The main findings obtained through RFM analysis and CLV modelling represent an essential guidance tool for shaping marketing strategies and supporting managerial decision-making:

- **Loyalty programs**: delivering exclusive benefits and tailored communication according to the tiered structure based on predicted CLV.
- **Customer retention**: identifying at-risk customers to develop targeted win-back campaigns by using personalized discounts and promotions.
- **Promotional strategies**: leveraging product category preferences to launch cross-selling and up-selling initiatives for increasing average order value.
- **Resource allocation**: concentrating marketing investments on high-value customers to ensure retention, while also targeting selected low-value segments to enhance customer spending.
- **Customer acquisition**: identifying potential profiles similar to those of existing high-value customers in order to establish long-term relationships.

CONCLUSIONS

Data-Driven Approach

The marketing strategy adopts a predictive, customer-centric model. By allocating resources based on individual Customer Lifetime Value (CLV), business efforts translate into long-term investment for specific segments.

Key Benefits

- Focus retention efforts on high-value customers who significantly contribute to overall revenue.
- Redirect resources from stable middle segments toward high-value or at-risk customers to improve the return on marketing investment.
- Use purchasing behaviour to provide more relevant messages and tailored promotions.

Next Steps

The next phase involves launching pilot programs for at-risk customers and offering exclusive rewards to top-tier customers. In particular, tracking performance over time will also be essential to analyse the impact of these initiatives on the overall revenue.

APPENDIX A (DATA OVEVIEW)

The **Fast-Moving Consumer Goods (FMCG) Customer Behaviour and Marketing Analytics dataset** is structured as a relational database composed of 14 interconnected tables related to multiple aspects of the purchasing process of an **Indian FMCG retailer**. Core aspects include customers, products, orders, stores, promotions, marketing campaigns, and customer support activities. The dataset spans from February 2023 to February 2026 and contains 356,242 customer orders across ten distinct FMCG product categories, collected from five different store formats operating in multiple cities and states in India.

Prior to database ingestion, **data quality assessment** was performed. The main issue concerned the promo_id field in the order_item table, where approximately 70% of values were missing. Since products are not always associated with an active promotion, these entries were stored as NULL values. Temporal variables were converted to appropriate date and time formats, whereas item-level discounts in the order_item table were capped at the corresponding line revenue, computed as unit price multiplied by quantity, to avoid negative transaction values. Moreover, a further issue involved **product pricing**. Since the price_change table stores prices effective from a specific year-week onward, 2.57% of rows in the order_item table had no matching price because the order date preceded the item's first recorded price. To preserve transactional data, these cases were assigned the closest subsequent price through a forward-matching approach, considered as a valid estimate of the actual price. Additional preprocessing included normalizing the relationship between the campaign_touch and marketing_channel tables by introducing a foreign key constraint and removing redundant information. No duplicate records or primary keys anomalies were detected. Ultimately, the cleaned dataset was successfully loaded into **PostgreSQL**, where structural and referential integrity are enforced by database constraints.

DATA DICTIONARY

Table Name	Records	Description
Category	10	Macro-classification of products
Product	200	Catalogue of available products with brand and SKU
Price change	1,978	Weekly history of price updates per product
Channel	4	Sales channels for purchasing
Store	140	Physical retail locations and types
External factor	21,840	Environmental and marketing conditions per store
Customer	10,000	Customer base profiles
Session	1,259,647	Digital activity and online behaviour of customers
Support ticket	1,600	Customer service logs with resolution and CSAT score
Marketing channel	12	Media platforms for advertising
Campaign touch	280,103	Marketing touchpoints per customer across channels
Promotion	280	Promotional campaigns with discounts and offers
Order	356,242	Transaction records of purchases
Order item	1,048,575	Specific item line within each order

MISSING VALUES ASSESSMENT

Table Name	Column	Missing Values	Percentage (%)	Interpretation
order_item	promo_id	732,376	69.84	Valid business condition (set as NULL)
All others	All columns	0	0	Complete records

SUMMARY STATISTICS (NUMERICAL VARIABLES)

Table: price_change Variable: unit_price	
mean	589.29
std	740.93
min	23.17
25%	128.45
50%	270.46
75%	890.68
max	4,415.46

Table: price_change		
	Variable: quantity	Variable: item_discount
mean	1.34	11.31
std	0.55	32.27
min	1	0
25%	1	0
50%	1	0.28
75%	2	6.76
max	3	971.36

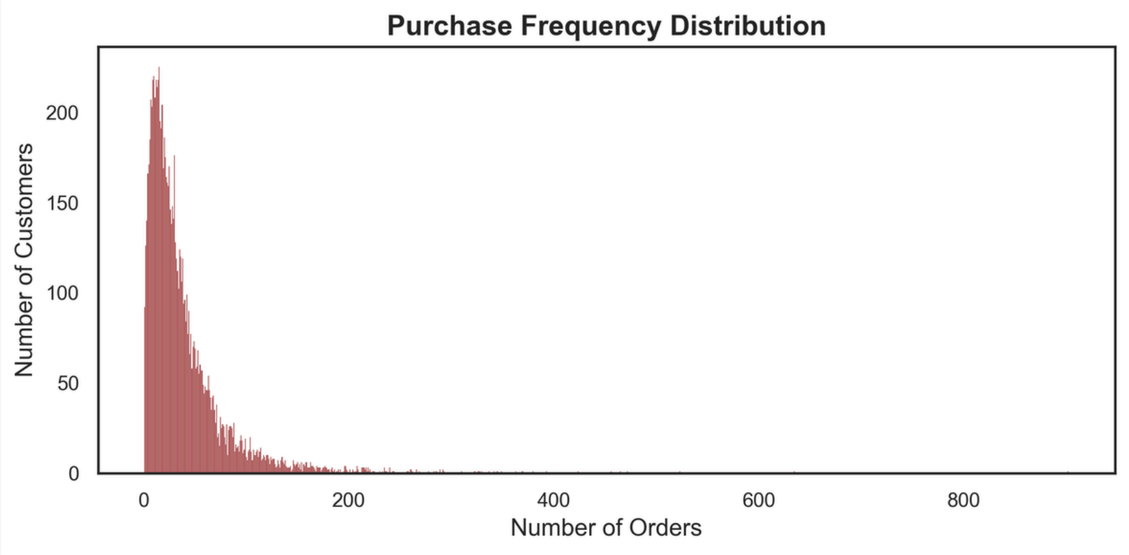
SUMMARY STATISTICS (CATEGORICAL VARIABLES)

Table: customer Variable: segment	
Value	25%
Mainstream	24%
Premium	18%
Health-focused	18%

Table: store Variable: store_type	
Supermarket	25%
Hypermarket	24%
General Trade	18%
Warehouse	18%
Modern Trade	15%

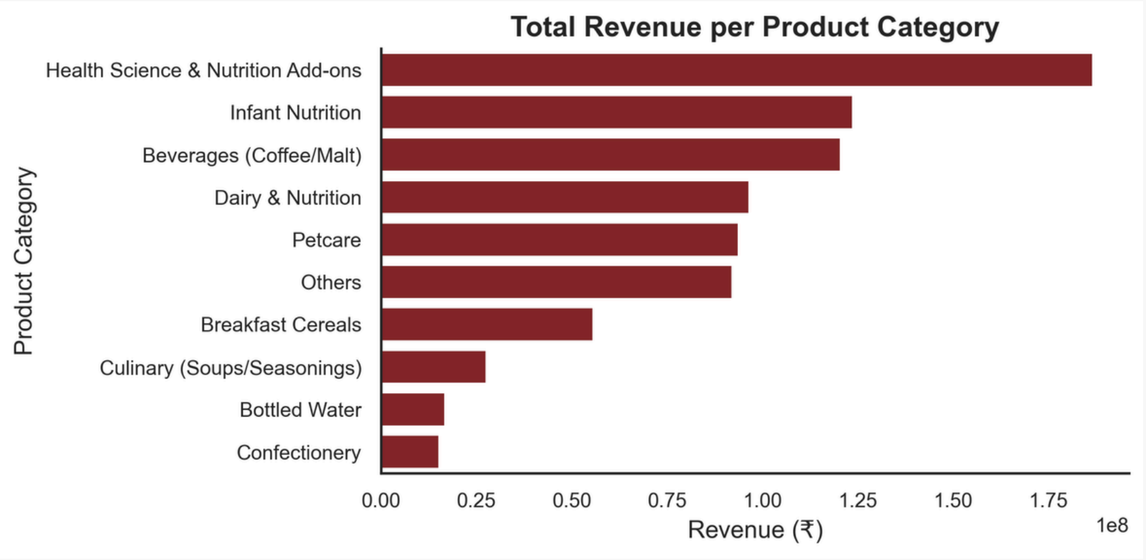
Table: category Variable: category_name	
Beverages	10%
Dairy & Nutrition	10%
Infant Nutrition	10%
Confectionary	10%
Culinary	10%
Breakfast Cereals	10%
Petcare	10%
Bottled Water	10%
Health Science & Nutrition Add-ons	10%
Others	10%

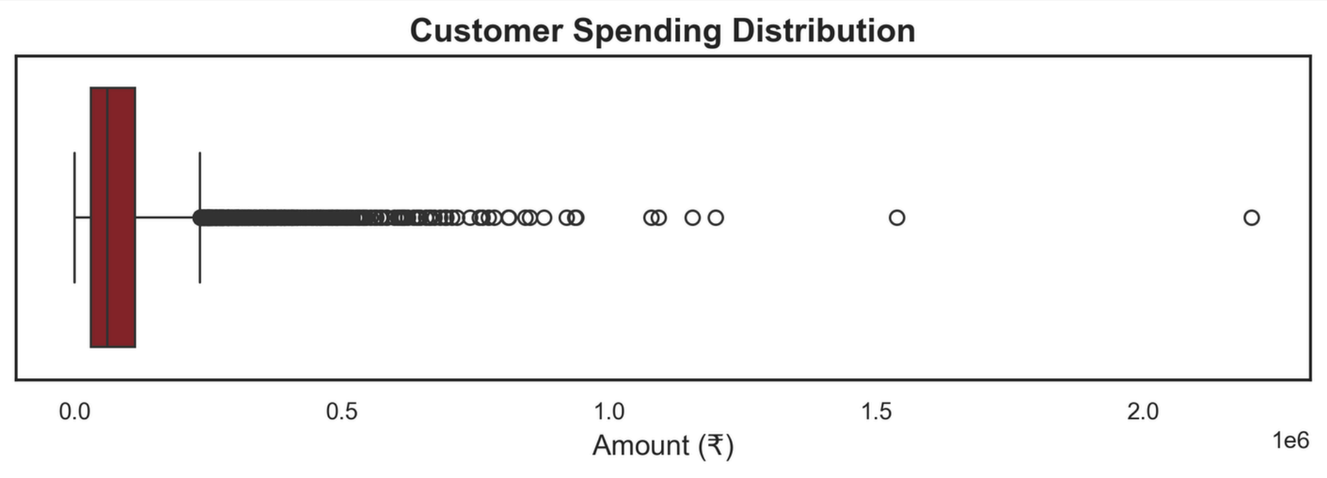
APPENDIX B (EXPLORATORY ANALYSIS)



During the analysed period, customers placed an average of 38 orders. However, the distribution is highly right-skewed, as the median is only 26 orders and 75% of the customers placed no more than 48 orders. While most customers show relatively low purchasing frequency, a small number of active customers significantly influence the average value. Indeed, the maximum observed value is 902 orders, corresponding to roughly 300 orders per year.

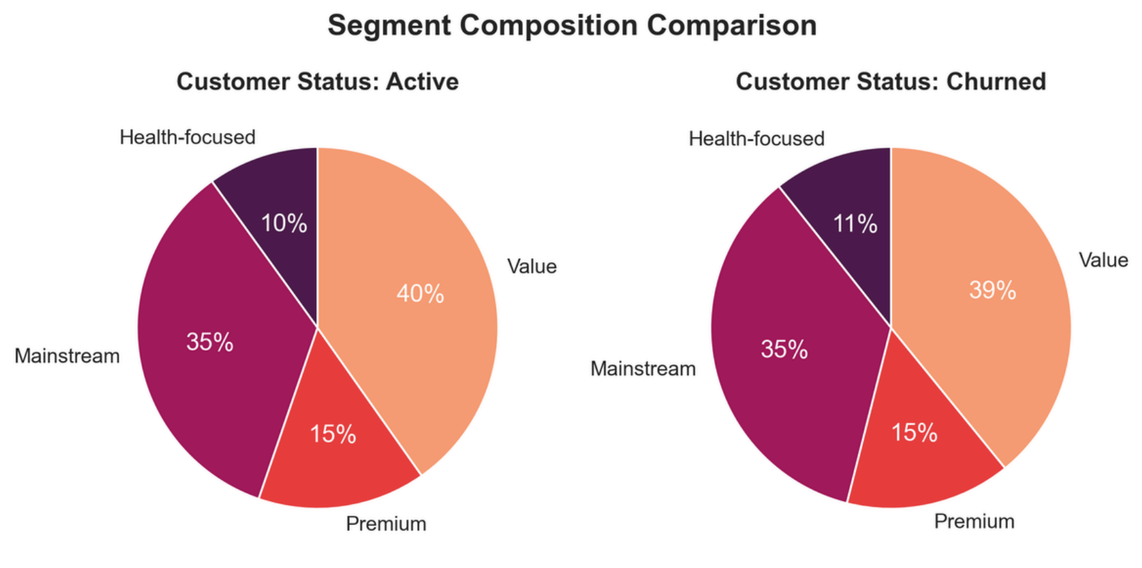
The Indian retailer operates across 10 FMCG product categories. In terms of total revenue, Health Science & Nutrition Add-ons represents the main category, generating approximately ₹ 185,000,000. It is followed by Infant Nutrition and Beverages. Therefore, these three categories account for the largest portion of total revenue, highlighting their importance within the company.



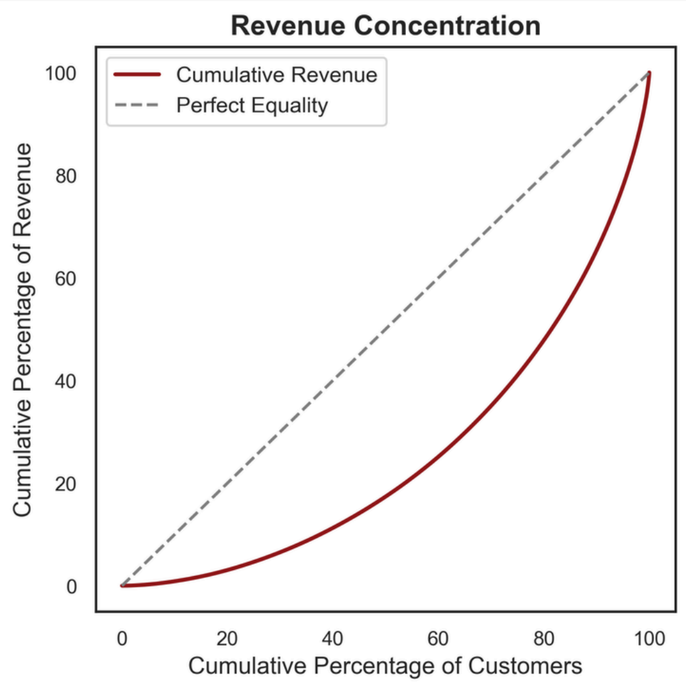


Customer spending ranges from a minimum of ₹ 34 to a maximum of ₹ 2,201,660. The average customer spending is ₹ 88,397, while the median is considerably lower at ₹ 60,713. Similarly to purchase frequency, the spending distribution is strongly right-skewed. In particular, most customers are concentrated in the lower spending ranges, with 75% spending less than ₹ 112,494. In contrast, a relatively small group of customers contributes to the upper tail of the distribution.

The comparison between active and churned customers does not reveal substantial differences in segment composition. The Value segment accounts for approximately 40% of customers in both groups, followed by the Mainstream segment at around 35%. Instead, Premium and Health-focused customers represent smaller shares, even though their proportions remain similar across active and churned customers.



The analysis of revenue concentration shows that the bottom 80% of customers account for approximately 50% of total revenue, while the remaining 20% generate the other half. Moreover, the cumulative curve lies below the line of perfect equality, indicating an uneven distribution of revenue across the customer base. In other words, a small portion of customers contributes to the largest share of global revenue.



SUMMARY OF KEY FINDINGS

- **Market development**: the company experienced strong growth throughout 2023, reaching peak revenue and order volumes at the end of the same year. More recent periods suggest a stabilisation at lower levels.
- **Customer behaviour**: both purchase frequency and customer spending exhibit strongly right-skewed distributions. In particular, most customers place relatively few orders and generate limited revenue, while a small group accounts for the largest share of sales.
- **Product categories**: Health Science & Nutrition Add-ons represents the most important source of revenue.
- **Customer segments**: the distribution of segments is very similar between active and churned customers, suggesting that no segment appears more exposed to churn than the others.
- **Revenue concentration**: revenue is not evenly distributed across customers, as the top 20% of customers generate approximately 50% of total revenue.

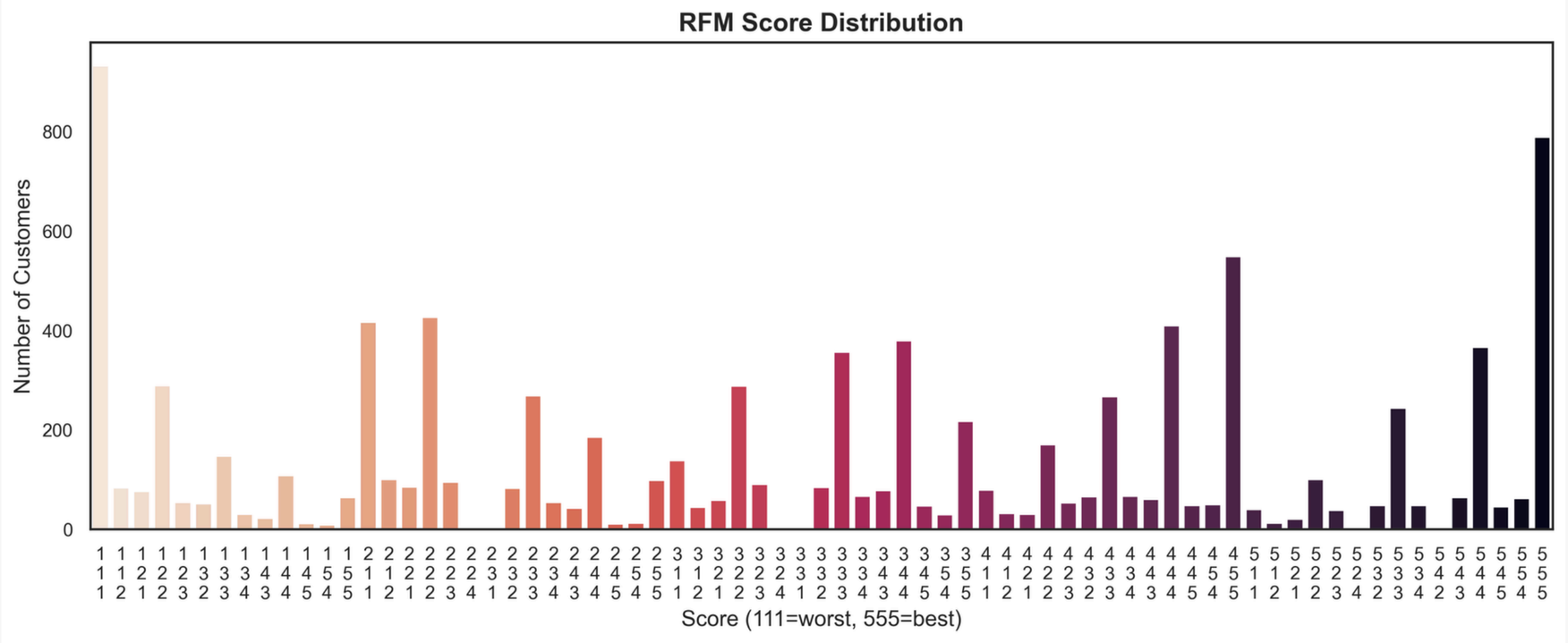
APPENDIX C (RFM ANALYSIS)

TOP 10 CUSTOMER TABLE BY RFM SCORE

Customer Id	RFM Score	Recency	Frequency	Monetary	Segment	Status	City	State
CUST000058	555	0	846	2,201,660.47	Premium	active	Kozhikode	Kerala
CUST002744	555	0	607	1,538,810.93	Value	active	Howrah	West Bengal
CUST008575	555	3	488	1,199,102.46	Premium	churned	Rajkot	Gujarat
CUST008014	555	5	437	1,156,252.93	Premium	active	Hubballi	Karnataka
CUST003831	555	0	444	1,092,418.85	Premium	active	Hyderabad	Telangana
CUST002370	555	0	429	1,079,149.96	Mainstream	churned	Panipat	Haryana
CUST008097	555	5	404	937,746.24	Value	active	Nashik	Maharashtra
CUST006740	555	0	366	935,973.45	Value	churned	Kanpur	Uttar Pradesh
CUST007473	555	4	349	920,531.40	Premium	churned	Mysuru	Karnataka
CUST007721	555	2	344	878,269.24	Mainstream	active	Panipat	Haryana

Considering the top 10 customers, all of them belong to the highest RFM scoring level (555). This means that they are recent, highly frequent, and high spending people, representing the largest portion of revenue of the Indian retailer. However, not all them are active: even top customers may disengage.

The distribution of RFM scores shows two dominant peaks, corresponding to the extreme groups (111 and 555). In the middle range, customer segments are more heterogeneous, with relatively similar frequencies across most combinations. However, there is a slight shift towards higher-value profiles, and some combinations are not present or only appear rarely. This pattern suggests that customer behaviour is concentrated in a few recurring groups, such as sporadic, low-engaged buyers or high-value segments.



APPENDIX D (CLV ANALYSIS)

TOP 20 CUSTOMER TABLE BY CLV

Considering the top 20 customers by Customer Lifetime Value, all of them are currently active customers. In particular, they exhibit high purchase frequencies, ranging from 246 to 846. Above all, their CLV values vary between ₹ 2.2 million and ₹ 6.8 million, surpassing the overall average of ₹ 272,425. Moreover, these customers belong to different customer segments and are geographically dispersed across several Indian states.

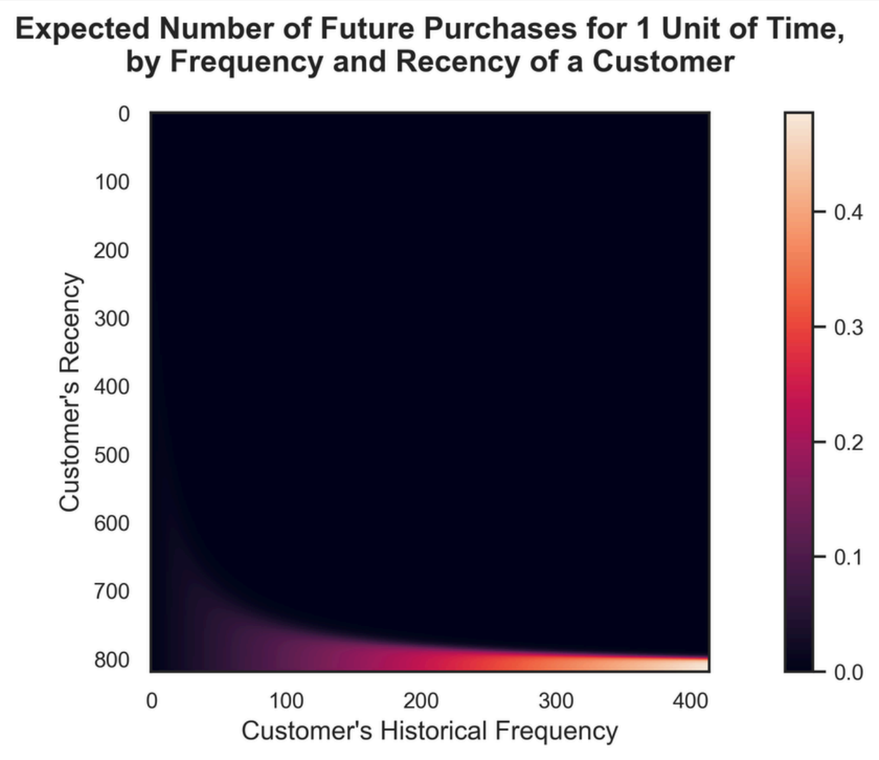
Customer Id	CLV	APV	PF	Segment	Status	City	State
CUST000058	6,785,137	2,602.44	846	Premium	Active	Kozhikode	Kerala
CUST002744	4,742,349.29	2,535.11	607	Value	Active	Howrah	West Bengal
CUST008575	3,695,426.51	2,457.18	488	Premium	Active	Rajkot	Gujarat
CUST008014	3,563,371.67	2,645.89	437	Premium	Active	Hubballi	Karnataka
CUST003831	3,366,646.06	2,460.40	444	Premium	Active	Hyderabad	Telangana
CUST002370	3,325,753.64	2,515.50	429	Mainstream	Active	Panipat	Haryana
CUST008097	2,889,971.82	2,321.15	404	Value	Active	Nashik	Maharashtra
CUST006740	2,884,508.38	2,557.30	366	Value	Active	Kanpur	Uttar Pradesh
CUST007473	2,836,918.66	2,637.63	349	Premium	Active	Mysuru	Karnataka
CUST007721	2,706,673.98	2,553.11	344	Mainstream	Active	Panipat	Haryana
CUST004846	2,623,375.15	2,411.45	353	Premium	Active	Varanasi	Uttar Pradesh
CUST003565	2,596,974.07	2,463.96	342	Value	Active	Howrah	West Bengal
CUST003896	2,502,867.36	2,545.89	319	Mainstream	Active	Pune	Maharashtra
CUST002154	2,499,246.83	2,526.36	321	Health-focused	Active	Nizamabad	Telangana
CUST005645	2,419,058.67	2,590.57	303	Value	Active	Bhubaneswar	Odisha
CUST002355	2,385,316.89	2,496.76	310	Mainstream	Active	Kota	Rajasthan
CUST006199	2,349,768.54	2,459.55	310	Mainstream	Active	Mangaluru	Karnataka
CUST005695	2,335,247.92	2,324.38	326	Mainstream	Active	Chandigarh	Punjab
CUST005657	2,280,161.87	2,341.37	316	Premium	Active	Faridabad	Haryana
CUST006807	2,203,347.88	2,906.29	246	Value	Active	Indore	Madhya Pradesh

APPENDIX E (CLV PREDICTION)

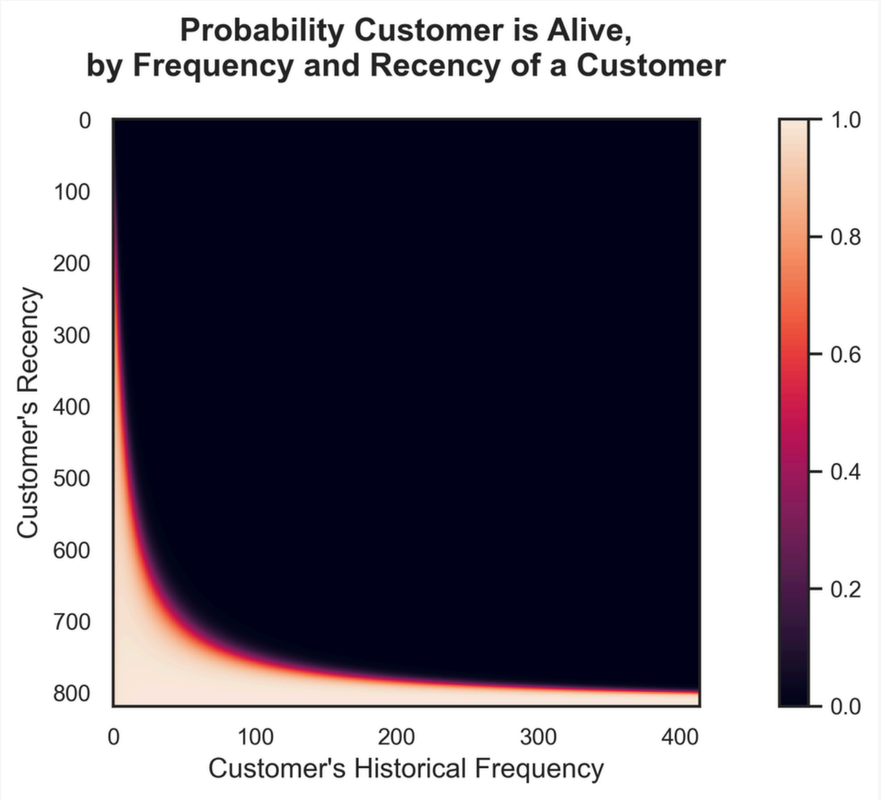
TOP 10 CUSTOMER TABLE BY PREDICTED CLV

Customer Id	Frequency	Recency	T	Monetary Value	Expected Frequency	P(alive)	Expected Avg Monetary	Predicted CLV
CUST000058	579	1,013	1,013	3,783.93	99.41	0.9999	3,785.66	363,528.96
CUST002744	436	913	913	3,520.86	82.8	0.9999	3,523	281,774.68
CUST003831	369	925	925	2,948.43	69.24	0.9999	2,950.55	197,352.30
CUST006740	289	773	773	3,153.27	64.47	0.9999	3,156.16	196,546.84
CUST008575	384	1,022	1,025	3,100.59	65.26	0.9997	3,102.73	195,576.32
CUST008014	347	1,030	1,035	3,329.79	58.43	0.9994	3,332.33	188,065.29
CUST002370	359	1,038	1,038	3,002.42	60.3	0.9999	3,004.64	175,004.27
CUST002154	264	770	772	3,063.57	58.98	0.9997	3,066.65	174,724.76
CUST004846	293	805	805	2,868.93	62.88	0.9999	2,871.53	174,399.89
CUST008097	322	941	946	2,881.87	59.14	0.9994	2,884.25	164,759.43

The analysis of the top 10 customers over the 6-month prediction horizon indicates that they exhibit strong performance both in historical behaviour and in expected future revenue. In particular, all of them have a very high probability of remaining active, and their predicted CLV ranges from approximately ₹ 165,000 to ₹ 364,000, well above the average value observed across the customer base.



The frequency-recency matrix suggests that customers characterized by both high purchase frequency and recent transactions have the greatest expected number of future purchases. Instead, the expected transaction rate decreases for customers whose last purchase occurred further in the past, even if their purchase frequency is relatively high. In addition, customers with fewer transactions but recent purchasing activity may have a high probability of also making future purchases.



The probability alive matrix indicates that customers with both high purchase frequency and recent transactions have the strongest probability of being active. In contrast, this probability decreases as the time since the last purchase increases, even among customers with a significant number of past orders. Therefore, recency emerges as a significant predictor of customer activity, as individuals with fewer purchases may still have a high probability of being active if they have purchased recently.

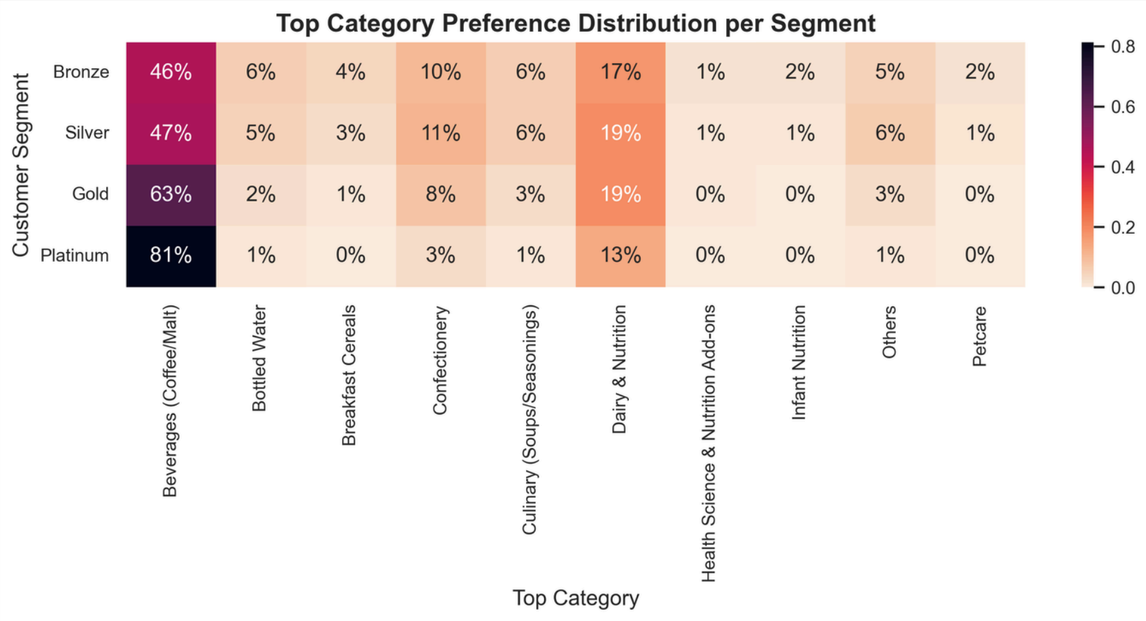
APPENDIX F (CUSTOMER SEGMENTATION)

SEGMENT SUMMARY TABLE

Customer Segment	Num Customers	Avg Spending	Avg Purchase Frequency	Avg P(alive)	Top Category	Top City	Top State	Top Customer Type
Bronze	1,876	2,280.81	16.18	0.18	Beverages (Coffee/Malt)	Hyderabad	Telangana	Value
Silver	2,813	2,466.44	13.32	0.94	Beverages (Coffee/Malt)	Kochi	Kerala	Value
Gold	2,813	2,631.93	31.57	0.99	Beverages (Coffee/Malt)	Hyderabad	Karnataka	Value
Platinum	1,876	2,673.33	82	1	Beverages (Coffee/Malt)	Dibrugarh	Assam	Mainstream

TOP PRODUCT CATEGORY PREFERENCES

The analysis of category preferences across customer segments highlights a structural difference between the categories that drive purchase frequency and those that generate the highest revenue. In particular, while Health Science & Nutrition Add-ons accounts for the largest portion of revenue, Beverages is the most frequently purchased category across all customer segments, followed by Dairy & Nutrition, which also represents a significant share of total purchases.



APPENDIX G (RECOMMENDATIONS)

The main findings obtained through RFM analysis and CLV modelling represent an essential **guidance tool for shaping marketing strategies and supporting managerial decision-making**. First, the ability to segment customers into Bronze, Silver, Gold, and Platinum levels based on their predicted Customer Lifetime Value for the next six months allows the business to develop effective **tiered loyalty programs**. In particular, exclusive benefits, personalized incentives, and tailored communication campaigns should target the small share of **high-value customers** that contribute significantly to overall revenue. Focusing on these customers can help improve retention rates, strengthen long-term relationships, and maximize future profitability. At the same time, the identification of **customers who are at risk of churning**, mainly because they have not purchased recently, provides an opportunity to implement targeted **win-back campaigns**. Although these individuals currently generate only a limited share of revenue, personalized discounts or promotional offers may encourage them to return and make further purchases.

In addition, customer segmentation based on recency, frequency, and monetary value allows the company to develop **communication and promotional strategies** tailored to the characteristics of each group. Therefore, customers can receive more relevant messages, product recommendations, and offers, increasing both engagement and loyalty. For example, high-value customers may benefit from exclusive rewards and personalized offers, while customers at risk of churning may be interested in discounts and short-term promotions. This personalized approach can improve customer satisfaction and strengthen relationships over time. Furthermore, the analysis of **top product categories** within each customer segment highlights Beverages as the main preference shared by the majority of customers. Although this category is not the largest source of overall revenue, it generates a high number of recurring purchases and helps maintain a high level of customer engagement. For this reason, the company should consider implementing **cross-selling campaigns** by proposing complementary products or services that contribute less to overall revenue, as well as **up-selling initiatives** aimed at promoting premium versions of existing products. These actions may increase average order value while also encouraging more heterogenous purchasing behaviours and improving higher customer satisfaction.

Regarding **resource allocation**, the company should mainly focus its marketing efforts on high-value and low-value customers. Retaining high-value customers is essential because they generate a large share of revenue. On the other hand, selected low-value customers could be targeted with specific campaigns aimed at increasing their engagement and spending. Customers in the middle segments appear to show more stable behaviour and may not require the same level of investment. Moreover, **customer acquisition** efforts should focus on attracting individuals with characteristics similar to those of existing high-value customers. The results of the RFM and CLV analyses show that the most valuable customers are those who purchase frequently and remain actively engaged over time. Therefore, rather than focusing only on increasing the number of new customers, the company should prioritize attracting customers who are likely to purchase regularly and build a long-term relationship. Ultimately, these insights can support the marketing analytics team in informing **managerial decisions** on loyalty programs, customer retention, promotional strategies, resource allocation, and customer acquisition.